

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF  
ECONOMICS Hypothesis Testing II

Week 11 · Two-sample, paired test and chi-square · Lectures 1 and 2

Tashkent International University · Faculty of Economics

2025/2026 academic  
year



## From one sample to two

### What did week 10 do

One sample. One mean or one proportion. Question: "Is  $\mu$  equal to such and such a number?"

And once you have  $\mu_0$ , the rest is mechanics: construct the null distribution, find the observed, date.

### What does week 11 ask

**"Are the two groups different?"**

This question is almost always encountered in practice: treatment and control, old and new, male and female, variant A and variant B.

This week,  $\mu_0$  means "the two groups are the same." If this is true, the label ("Horror"/"Comedy") means nothing — it can be confused.

This whole week comes from this one idea.

A common mistake And the most important question this week is not a statistical one, but a design question: are the observations paired or unpaired? A wrong answer reverses the conclusion — we'll see that today.

### Lecture 1 — Two-choice 1. Label Shuffling: A Randomization Test for Two Groups.

2. Difference of Two Means: The Welch  $t$ -Test and Why It's Standard.

3. Paired Test: When and Why SEs Get Too Small.

4. Difference of Two Proportions; For a  $2 \times 2$  Table,  $\chi^2 = z^2$ .

### Lecture 2 — Categorical Data and Multiple Testing 5. $\chi^2$ -Conformity Test — With Benford's Law.

6.  $\chi^2$ -Independence Test; Expected Frequencies and Conditions.

7. Cramer's  $\phi$  — How Strong Is the Question After  $\chi^2$ ?

8. Multiple comparisons: Bonferroni, Holm, FDR — the promise of week 10.

Prerequisites CLO7. Formulate a hypothesis, select an appropriate test, conduct it, and interpret the results with effect size, confidence interval, and  $p$ . Bloom: Apply → Analyze → Evaluate. New part this week — “choose”.

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## Mixing the label

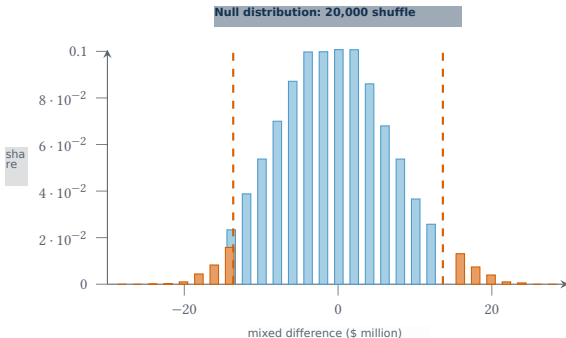
## Randomization test: Label under $\mu_0$ is meaningless

Question Are Horror and Comedy budgets different? Sample: 15 films from each genre (LCG seed 20261101).

Conditions  $\mu_H = 19,000$ ,  $\mu_C = 17,747$ ;  $\sigma_H = 32,633$ ,  $\sigma_C = 20,739$  (million \$). Difference  $\mu_{\text{fall}} = -13,633$ .

### Procedure (F153)

$\mu_0$ : label is irrelevant. So put 30 values in one box  $\rightarrow$  shuffle  $\rightarrow$  call 15 random ones "Horror"  $\rightarrow$  write the difference. Repeat 20,000 times.



### Result

$|\text{difference}| \geq 13.633$ :  $1,277/20,000 \Rightarrow \mu_{\text{rand}} = 0.063850$ . Welch  $\mu$ -test  $\mu = 0.063468$  — difference only 0.60%: two completely different paths, same answer.

## Same question, more information — Week 10 lesson returns

With  $\alpha = 15$   $\hat{\mu} = -13.633$   $\hat{\sigma} = 0.0639$   
95 % CI:  $[-28.085; 0.819]$  —  
within zero

Decision:  $H_0$  cannot be rejected.

SE = 1.785, df = 385.56,  $\hat{\mu} = -4.327$   $\hat{\sigma} = 1.93 \times 10^{-5}$   
95 % CI:  $[-11.234; -4.215]$

It was not visible at  $\alpha = 15$ . It was clearly visible at  $\alpha = 811$ . “Non-significant” never means “no difference.” This is exactly the power problem from week 10.

Question to the class If we had stopped at  $\alpha = 15$  and declared that “genres do not differ” — what kind of error would we have made? And who would have noticed this error?

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**Two independent choices**

## Difference of two means: where does the standard error come from

From week 6  $\text{Var}(\bar{x} - \bar{y}) = \text{Var}(\bar{x}) + \text{Var}(\bar{y})$  — add, not subtract:

$$\text{SE}(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

### Welch df (F130) and statistics (F131)

$$\nu = \frac{\left( \frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}, \quad t = \frac{\bar{x}_1 - \bar{x}_2 - \delta_0}{\text{SE}}$$

$\nu$  is not an integer — this is normal.

### Interval (F132) and effect (F135)

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, \nu} \text{SE} = (\bar{x}_1 - \bar{x}_2) \pm$$

### Wheat yield, 1992-2018 (t/ha)

|            | $n$ | $\bar{x}$ | $s$      |
|------------|-----|-----------|----------|
| Uzbekistan | 27  | 3.604815  | 1.179486 |
| Kazakhstan | 27  | 1.026741  | 0.259923 |

Difference: 2.578074 t/ha  
Variance ratio: 20.59

### Two counts — same, not

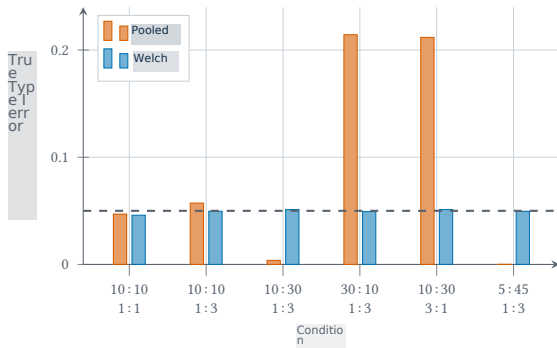
SE df

Welch 0.232439 28.52  $7.39 \times 10^{-12}$   
Pooled 0.232439 52  $2.58 \times 10^{-15}$

Since  $\nu_1 = \nu_2$ ,  $\nu = 11.0914$  is exactly the same. Only the df is different — and pooled  $\nu$  is 2,870 times smaller.

95 % CI (Welch): [2.1023; 3.0538] · Cohen  $d = 3.019$ .

## Why Welch is a measured response, not a belief



Conditions Each column: 20,000 replicates,  $\sigma_0$  true,  $\alpha = 0.05$  (dashed line). Labels:  $\sigma_1:\sigma_2$  and  $\sigma_1:\sigma_2$ .

### Pooled breaks down in both directions

The variance in the small group is large: the Type I error is 0.2143 — 4.3 times larger than promised. The test does not do what it promises.

The variance in the small group is small: 0.0001 — the test almost never rejects, i.e. loses power. This is also an error, but it is not noticeable.

### Welch range: [0.0459; 0.0512]

Around 0.05 in all six cases. Rule: always Welch. It is not necessary to pre-test the variances to choose it — on the contrary, pre-testing itself creates the problem of multiple comparisons (Lecture 2).

## Stop: the calculation is correct, the condition is violated

Hidden flaw in the wheat example These two series are a time series. The independence condition says "each observation is independent of the other". Are consecutive years independent?

Lag-1 autocorrelation: Uzbekistan:  
 $0.9658$  limit  $\pm 0.975/\sqrt{27} = 0.3772$   
Kazakhstan:  $-0.0003$

✗ The condition is violated in the Uzbekistan series.

What to do The violation is only in one series: Uzbekistan has a strong growth trend, while Kazakhstan's yield is weather-dependent and trendless.

A violation of one series is enough: the  $\alpha$ -value is not valid in the formal sense.

The effect is so large ( $\alpha = 3.02$ ) that the conclusion is almost certainly unchanged — but this cannot be said based on the  $\alpha$ -value.

### Audience Question

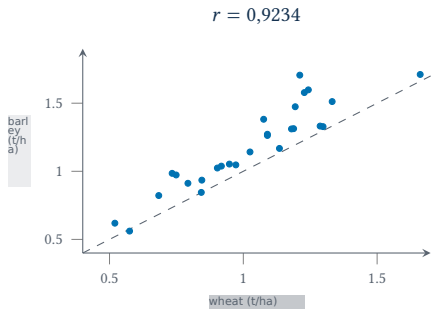
This slide is intentional. The most common mistake in statistics is not choosing the wrong formula, but applying the right formula to data that does not meet the condition. After each test: "Are the conditions met?"

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## Paired test

## Key fact of the week: same data, opposite conclusion

Real data Kazakhstan, wheat and barley yields, 1992-2018. Each year — one pair: same soil, weather and policy.  $n = 27$  pairs;  $\bar{d} = -0.154889$  t/ha (barley higher).



95 % CI:  $[-0.2025; -0.1073]$  — outside the null

✓  $H_0$  is rejected

### Analysis 2: assuming it is independent (error)

The difference estimate is exactly the same:  
 $-0.154889$  SE =  $0.077458$  — 3.34 times larger  $t = -1.9996$ , df = 50.61,  $p = 0.050920$  95% CI:  $[-0.3104; 0.0006]$  — inside the null

✗  $H_0$  is not rejected

### Audience question

Same numbers. Same difference. Opposite decision. The difference is in the answer to only one question: are the observations even?

# Why pairing works — it's a fact, not magic

## Paired difference variance (F138)

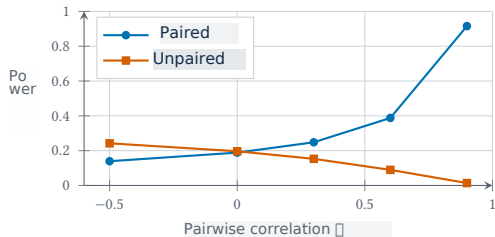
$$\text{Var}(\bar{d}) = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$$

This is the full form of  $\text{Var}(\bar{d})$  from week 6 ( $\rho = 0$ ).

## Checking with numbers

$0.067560 + 0.094433 - 2(0.923414)(0.259923)(0.307300) = 0.014479$ ; observed  $\sigma^2 \bar{d} = 0.014479$ , difference  $2.95 \times 10^{-17}$ . Same.

Definition At  $\rho = 0.9234$ , the variance is reduced by a factor of 11.19: pairing eliminates the overall effect of year.

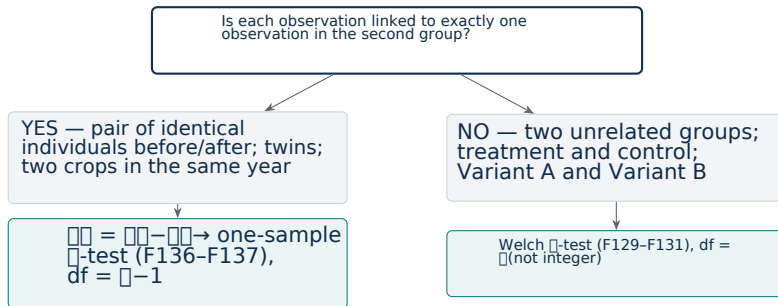


Conditions Simulation:  $n = 27$  pairs, true difference 0.30 SD, 20,000 replicates,  $\alpha = 0.05$ .

## Two-sided warning

$\rho = 0.9$ : paired power 0.9156 vs. unpaired 0.0138. The reverse at  $\rho = -0.5$ : 0.1394 vs. 0.2422. But this is not a choice: if the design is paired, paired analysis is mandatory.

## Which test? — answer the design question



### Common error

**Most common error: counting  $n$  twice in paired data. 27 pairs =  $n=27$ , not  $n=54$ ; df = 26.**

### Audience Question

The test is not chosen after the data. The design is known before the data are collected — so the test is known beforehand. This is the idea behind the pre-recording in Week 10.

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**Two shares**

## Difference between two shares: test from $\pi_0$ , intermediate from $\hat{\pi}$

Real question “Breakthrough” movie = worldwide gross > 2.5× budget (industry rule). Is PG-13 and R different?

useful total  $\hat{\pi}$

|          |          |          |   |
|----------|----------|----------|---|
| PG-13    | 358 896  | 0.399554 | R |
| 409 1113 | 0.367475 |          |   |

Difference:  
0.032078

### Two standard errors (F141-F144)

$$\hat{\pi}_{\text{bir}} = \frac{\pi_1}{n_1} + \frac{\pi_2}{n_2} = 0.381782$$

$$SE_0 = \sqrt{\hat{\pi}_{\text{bir}}(1 - \hat{\pi}_{\text{bir}})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} = 0.021806$$

$$SE_{IO} = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} = 0.021831$$

### Most common technical error of the week

The test assumes that  $\pi_0$  is true ( $\pi_1 = \pi_2$ ) — so the total share is estimated by combining:  $SE_0$ . The confidence interval does not assume  $\pi_0$  and each group uses its own  $\hat{\pi}$

Here the two SEs are almost equal (0.000025 difference), but when  $\hat{\pi}_1$  and  $\hat{\pi}_2$  are very different, they diverge.

### Result

$$z = \frac{0.032078}{0.021806} = 1.471107 \quad p = 0.141262$$

95% CI: [-0.010710; 0.074866] — within zero

Test and interval are the same conclusion:  $\pi_0$  cannot be rejected. (The binary rule from week 10.)

$\pi_1 + \pi_2 = 2,009$  — a large sample. A large  $\pi$  does not make any difference “significant”; it only helps to see the difference that exists. There is nothing to see here.

## 2 × 2 table: $\chi^2 = \chi^2$ — not a coincidence

### Same data, two different symbols

|       | useful | useless | total |
|-------|--------|---------|-------|
| PG-13 | 358    | 538     | 896   |
| R     | 409    | 704     | 1113  |
| total | 767    | 1242    | 2009  |

Difference:  $4.44 \times 10^{-15}$  — computer accuracy.

### F152

For a 2 × 2 table, the  $\chi^2$  and the two-tailed  $z$ -test are the same test, just written differently.

So why do you need both?  $\chi^2$ -test gives direction (+ or −) and can be one-sided.  $\chi^2$  is never one-sided — squaring loses direction.  $z$ -test gives a confidence interval for the difference.  $\chi^2$

does not. Therefore, prefer the  $z$ -test on 2 × 2.  $\chi^2$  does generalize to tables larger than 2 × 2 — the  $z$ -test does not.

### Effect size — don't leave $\chi^2$ alone

Difference: +0.032078 (absolute) Relative: +8.73% Relative risk RR = 1.0873 Odds ratio OR = 1.1454 (F145)

The absolute difference is the most understandable measure for proportions. OR is the most misleading: 1.15 does not mean “15% more.”

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## Chi-square

## $\chi^2$ statistics: one idea, two applications

### General form (F146)

$$\chi^2 = \sum_{\text{all cells}} \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

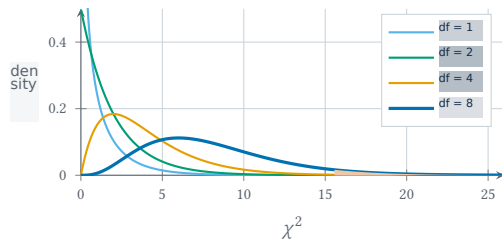
Each cell contributes.

### The difference is only in the "expected"

Fit (F147): from the expected theory;  $\chi^2 = \chi^2(0)$  ,  
 $df = n - 1 - (\text{estimated parameters})$ .

Independence (F148): from the expected data itself;  $\chi^2 = (\text{row total})(\text{column total})/n$ ,  $df = (n-1)(n-1)$  (F149).

Condition Each expected frequency  $\geq 5$  (not observed — expected). If violated: combine categories or exact test.



### Conditions

Shaded area: upper 5% for  $df = 8$ ; critical value  $\chi^2_{0.95}(8) = 15.5073$ .

### Three properties

1.  $\chi^2 \geq 0$  — sum of squares. 2. Rejection is always in the right tail: small  $\chi^2$  means "good fit". 3.  $E(\chi^2) = df$  — we will measure in the next slide.

## Consistency test: Benford's law — one file, two answers

### Benford's law

A one-digit number in natural numbers spanning several orders of magnitude:  $P(d = \text{first number}) = \log_{10}(1 + \frac{1}{d})$

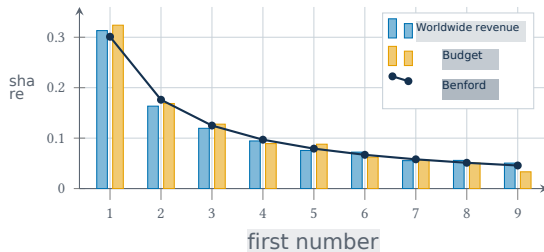
$P(d=1) = 0.3010$ ,  $P(d=9) = 0.0458$ . Used in auditing to detect fraud.

### Same file, same test

Column 2(df 8)

|                 |         |        |
|-----------------|---------|--------|
| World income    | 8.1242  | 0.4214 |
| Budget          | 18.4609 | 0.0180 |
| Domestic income | 32.3800 | 0.0001 |

Income is consistent. Budget is inconsistent.



### The reason is not in the statistics — the origin of the data

84.96% of budgets are whole multiples of 1 million, 50.43% are whole multiples of 5 million; only 267 different values for 2,467 films. In worldwide revenue, only 1.34% are whole multiples of 1 million, the number of different values is 2,436.

Budget is the announced and rounded number; revenue is the measured number.

# Test of Independence: Expected Frequencies and Condition

## Observed: Rating × Genre (2,458 Movies)

Action Advent. Comedy Drama Horror

G 0 41 3 5 0 PG 6 246 39 106 3 PG-13 178 62 266 339  
51 R 174 14 292 479 154

## Raw Table: Condition VIOLATED

$\chi^2 = 1159.667$ ,  $df = 12$ ,  $p < 10^{-240}$  — but the smallest expected frequency = 4.1465 (1 in 20 cells is less than 5). There are only 49 movies in row G: the category is very sparse.

## Correction: We combine G and PG

$\chi^2 = 1141.736$ ,  $df = 8$ , smallest expected = 37.9951 — the conclusion has not changed, but it is now justified.

## Cramer $\phi$ (F151): question after $\chi^2$

$$V = \sqrt{\frac{\chi^2}{n(\min(r, c) - 1)}} = \sqrt{\frac{1141.736}{2458 \cdot 2}} = 0.4819$$

$\phi \approx 0.1$  weak ·  $0.3$  moderate ·  $0.5$  strong — here strong.  $\phi^2$  grows with  $\chi^2$ ,  $\phi$  does not.

## Where? — residuals (F150)

$\chi^2$  just says “there is a relationship”; where  $\text{cell} = (\text{observed} - \text{expected}) / \sqrt{\text{expected}}$  shows.

cell observed residual

G/PG × Adventure 287 +27.1019 R  
× Adventure 14 -11.7287 G/PG ×  
Action 6 -7.3448

Source single cell: family adventure movies.

## Where does the $\chi^2$ distribution come from? — let's measure

Let's set Experiment 0 to true: the row and column are truly uncorrelated.  $n = 300$  observations,  $3 \times 5$  table, 20,000 replicates (LCG, seed 20261400). We calculate  $\chi^2$  each time.

### Result — comparison with theory

#### simulation theory

mean 8.0353 df = 8 variance  
15.6764  $2 \cdot \text{df} = 16$  95% quantile  
15.4946 15.5073  $P(\chi^2 > \text{crit})$   
0.0498 0.05

### Why is this important

**The  $\chi^2$  distribution does not fall from the sky: it is the sum of squares of independent normal quantities (week 5, MLT).**

Degrees of freedom The answer to the question “how many cells are free”: When the row and column totals are recorded in a  $3 \times 5$  table, only  $(3 - 1)(5 - 1) = 8$  cells are free.

### Approximation — not exact

The  $\chi^2$  distribution is an approximation. It only works when  $n$  is large and the expected frequencies are large enough. If the expected frequencies are small, the true distribution is discrete and the  $\chi^2$  curve does not fit it — so we have the condition  $n \geq 5$ . This is exactly the “normal approximation” problem from Week 7.

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## Multiple comparisons

## Promise in week 10: not one, but ten tests

Real problem 5 genres  $\Rightarrow$  all pairs =  $\binom{5}{2}$  = 10 tests. Question: do genres differ in their utility share?

| genre     | $n$ | utility share |        |
|-----------|-----|---------------|--------|
| Horror    | 210 | 117           | 0.5571 |
| Adventure | 363 | 167           | 0.4601 |
| Action    | 360 | 144           | 0.4000 |
| Comedy    | 601 | 227           | 0.3777 |
| Drama     | 933 | 304           | 0.3258 |

Three corrections Bonferroni (F155):  $\alpha^* = \min(\alpha, 1)$ . Simple, conservative. Guarantee:  $\text{FWER} \leq \alpha$ .

**Holm (F156):** Sort the  $\alpha$ s; multiply by  $(n - i + 1)$  to the  $i$ -th; make monotonic. Exactly the same guarantee, but with more power. Always Holm instead of Bonferroni.

Benjamini-Hochberg (F157): multiply by  $\alpha/n$  to the  $i$ -th. Guarantees something else:  $\text{FDR} \leq \alpha$  (F158) — the fraction of false positives among the findings, not “at least one false positive”.

From week 10 (F154)  $\alpha = 0$  is true everywhere, but  $P(\text{at least one significant}) = 1 - 0.95^{10} = 0.4013$ .

That is: even if there is no difference, at least one of the 10 tests is more than 40% likely to be “significant”.

### Audience Question

They answer different questions. The right choice is not a question of “which one is more rigorous”, but rather a question of “how much will the error cost me”.

# The correction changes the conclusion — in ten pairs

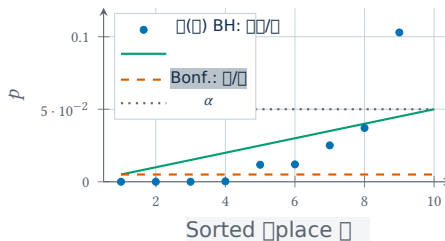
pairs  $\square$  Bonf. Holm BH

Drama-Horror 0.000000 0.000000 0.000000 0.000000 Comedy-Horror  
0.000006 0.000059 0.000053 0.000021 Advent.-Drama 0.000006  
0.000064 0.000053 0.000021 Action-Horror 0.000281 0.002808  
0.001966 0.000702

Advent.-Comedy 0.011733 0.117329 0.070397 0.019999  
Action-Drama 0.011999 0.119992 0.070397 0.019999 Advent.-Horror  
0.025108 0.251081 0.100432 0.035869 Comedy-Drama 0.037100  
0.370999 0.111500 0.046375

Action-Advent. 0.102937 1.000000 0.205874 0.114375 Action-Comedy  
0.491972 1.000000 0.491972 0.491972

Significant at  $\alpha = 0.05$ : uncorrected 8/10 · Bonferroni 4/10 · Holm 4/10 · BH 8/10



## Reading the graph

BH: Find the largest  $\square$  for which  $\square(\square) \leq \square/\square$  — reject all up to that point. Bonferroni: those below the horizontal line. The BH line is sloping, so it finds more.

Conditions For the same 10 pairs of budgets: 8/10 without adjustment, 8/10 with Bonferroni. Adjustment is not a penalty.

## What do the three methods guarantee — with simulation

$\pi = 5$  (i.e. 5 out of 20 have a real effect):

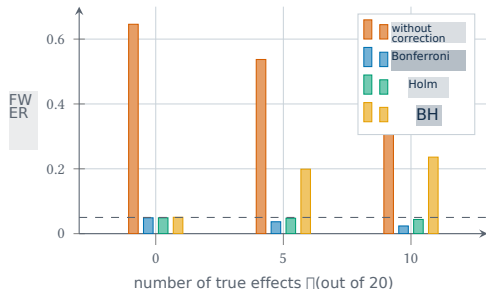
method FWER FDR power

|             |        |        |        |
|-------------|--------|--------|--------|
| uncorrected | 0.5371 | 0.1150 | 0.9879 |
| Bonferroni  | 0.0367 | 0.0075 | 0.8424 |
| Holm        | 0.0475 | 0.0095 | 0.8584 |
| BH (FDR)    | 0.1988 | 0.0377 | 0.9497 |

Three conclusions 1. Bonferroni and Holm capture FWER below 0.05 — but Holm does so with more power (0.8584 and 0.8424). Holm is always preferable.

2. BH does not capture FWER (0.1988) — and this is not a mistake. It captures the FDR:  $0.0377 \leq 0.05$ .

3. Without correction: Even when  $\pi_0$  is true everywhere ( $\pi = 0$ ), the false discovery probability is 0.6457.



Conditions  $\pi = 20$  tests,  $n = 30$  each, effect 0.80 SD, 10,000 replicates. Dashed line = 0.05.

## Statistical ethics: “choosing a test after the data”

This week's unique abuse  $\alpha$ -hacking took three forms in week 10: multiple testing, arbitrary stopping, and clustering.

A fourth form appears this week: choosing a test based on the outcome.

- ▶ Welch gave  $\alpha = 0.06 \rightarrow$  try pooled
- ▶ Paired didn't work  $\rightarrow$  consider it uncorrelated
- ▶  $\alpha/2$  insignificant  $\rightarrow$  combine categories “conveniently”
- ▶ Adjusted  $\alpha$  large  $\rightarrow$  drop the adjustment

Why it's especially dangerous Each of these seems justified when taken individually. “The variances seemed equal.” “The category was too sparse.” Nobody lies.

But if the decision is made after seeing the result, the  $\alpha$ -value loses its meaning — because it is built on the assumption of “one predetermined test.”

### Defense — three habits

1. Choose the test from the design, not the result. The design is known before the data.
2. Count and publish how many tests you have done — hidden tests are the most harmful.
3. Show both the uncorrected result and the corrected one. Transparency, not hiding, is the solution.

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**Summar  
y**

## Common Student Mistakes and Misconceptions

1-5 1. Treating paired data as independent. In today's example, this raised  $\chi^2$  from  $4.3 \times 10^{-7}$  to 0.0509—the conclusion was reversed.

2. Counting  $\chi^2$  twice in paired data. 27 pairs =  $\chi^2=27$ , not  $\chi^2=54$ ;  $df = 26$ .

3. Using the same SE for the IO and test in a proportion test. The test  $\chi^2$  is one, the interval is each group's own  $\chi^2$ .

4. Comparing the observed frequency with 5. The condition is about the expected frequency.

5. Trying to make  $\chi^2$  one-sided. Squaring loses direction. If direction is needed —  $\chi^2$ -test.

6-10 6. " $\chi^2$  significant  $\Rightarrow$  strong relationship". It increases with  $\chi^2$ . Strength — Cramer  $\chi^2$ .

7. " $\chi^2$  shows cause and effect". It doesn't even show which cell it is — you need the residuals for that.

8. Check variances first and then choose a test. That's multiple comparisons. Always Welch.

9. "Adjustment destroys the result". In the budget example, adjustment left 8/10 as 8/10. It only affects the marginal results.

10. " $\chi^2 > 0.05 \Rightarrow$  groups are equal". At  $\chi^2 = 15$ ,  $p = 0.064$ ; at  $\chi^2 = 811$ ,  $p = 0.0000193$ . The effect did not change.

## Summary: one table, three sentences

### Which test when

#### Condition Test Effect

Two independent groups, number Welch  $\chi^2$   
Cohen  $d$

Paired observations, number Paired  $\chi^2$   $\chi^2$

Two-proportion  $\chi^2$ -test ( $\chi^2$ one) difference, RR, OR

One categorical,  $\chi^2$  fit with theory —

Two categorical  $\chi^2$  independence Cramer  $\phi$

Conditions are questionable,  $\chi^2$  small Randomization is the difference itself

### In three sentences

1. The design chooses the test, not the data. Paired or independent — this question must be known before the answer is calculated.

2.  $\chi^2$ —one idea ( $\sum(\chi^2 - E)^2 / E$ ), two “expected” ones. And it only says there is a relationship; the strength is  $\phi$ , and the residuals tell the difference.

3. If you do multiple tests, you need to make adjustments—and which adjustments you need to make depends on the question “how much will the error cost me,” not on statistics.

### Next Steps

This week we compared groups. In Week 12, we model the form of the relationship between variables: simple linear regression. Cramer answered the question “how strong”  $\phi$  is—regression answers the question “what kind.”

## References

### Main reading

- ▶ Çetinkaya-Rundel M., Hardin J. Introduction to Modern Statistics (IMS, 2nd ed.), Chapters 20–23 (summary for two-choice and categorical data). Open source: [openintro.org/book/ims](https://openintro.org/book/ims)
- ▶ Newbold P., Carlson W., Thorne B. Statistics for Business and Economics (9th ed.), Chapter 10 and 11.1–11.3.
- ▶ Heumann C., Schomaker M., Shalabh. Introduction to Statistics and Data Analysis, Chapter 10 (2 tests).

### This week's required reading

- ▶ Delacre M., Lakens D., Leys C. (2017). Why psychologists should by default use Welch's  $t$ -test instead of Student's  $t$ -test. *International Review of Social Psychology* 30(1), 92–101. — the source of today's simulation.
- ▶ Benjamini Y., Hochberg Y. (1995). Controlling the false discovery rate. *Journal of the Royal Statistical Society B* 57(1), 289–300. — The article that gave birth to the idea of FDR (first 4 pages).

### Supplement

- ▶ Holm S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics* 6(2), 65–70.
- ▶ Nigrini M. (2012). *Benford's Law: Applications for Forensic Accounting, Auditing, and Fraud Detection*. Wiley.
- ▶ Wasserstein R., Lazar N. (2016). The ASA statement on  $p$ -values — from week 10, still applies.